Lecture Notes: Introduction to Data Science

Sample notes | Data Science | MBU DS LAB

1. What is Data Science?

Data Science combines statistics, programming, domain knowledge, and analytical methods to extract useful insights from data.

2. Typical workflow

- Define the problem and success criteria.
- Collect data from reliable sources.
- Clean missing, duplicated, or inconsistent values.
- Explore and visualize patterns.
- Build and evaluate a suitable model.
- Communicate findings and limitations.

3. Common data types

- Numerical: age, temperature, price.
- Categorical: department, product type.
- Time series: daily sales or sensor readings.
- Text: reviews, messages, documents.

4. Important reminder

A model's quality depends on data quality, appropriate evaluation, and avoiding data leakage. Correlation alone does not establish causation.

Sample educational material for the MBU DS LAB Resources page. Verify course-specific requirements with your faculty.